11.1 A single phase, 240/120 V transformer has the following parameters:

$$R_1 = 1\Omega$$
$$R_2 = 0.5\Omega$$
$$X_1 = 6\Omega$$
$$X_2 = 2\Omega$$
$$R_o = 500\Omega$$
$$X_o = 1.5\text{k}\Omega.$$

A load of 10Ω at 0.8 power factor lagging is connected across the low-voltage terminal of the transformer. The voltage across the primary side is 240^{V}. Compute:

a) The current at the secondary side.

First, referencing the schematic in the midterm review lecture slides (page 22), we'll need to find the value of the secondary-side impedances when reflected to the primary side; in other words, how much impedance is the primary side "seeing" from R_2, X_2, and the load impedance? Using the formula $Z'_{\text{load}} = Z_{\text{load}} \left(\frac{N_1}{N_2}\right)^2$, we can find the reflected impedances. Starting with $R_2 + jX_2$, $(0.5 + j2)\left(\frac{240}{120}\right)^2 = 2 + j8\Omega$. Then, the load impedance may first be converted to complex form by finding the phase angle as a function of the power factor, where $\theta = \cos^{-1}(0.8) = 38.87°$, and so $Z_{\text{load}} = 10\cos(38.87°) + j10\sin(38.87°) = 8 + j6\Omega$, and reflecting this impedance using the previous formula, $Z'_{\text{load}} = 4(8 + j6) = 32 + j24\Omega$.

Now that the impedance is properly reflected, we can add the primary side impedances to the mix, $1 + j6\Omega$, for a total impedance "seen" by the primary side of $Z_{\text{total}} = 34 + j38 = 51\angle 48.18°\Omega$. Finally, finding reflected current, $I'_2 = \frac{V_1}{Z_{\text{total}}} = \frac{240\angle 0°}{51\angle 48.18°} = 4.71\angle{-48.18°}$. Then, reflecting the current back to the secondary side, $I_2 = I'_2\left(\frac{240}{120}\right) = 9.42\angle{-48.18°}$.

b) The voltage at the secondary side.

A little easier than the last one; knowing that voltage is current times impedance, we may take the secondary side current and multiply it by the load impedance to find the voltage as $V_2 = I_2 \cdot Z_{\text{load}} = 9.42\angle{-48.18°} \cdot 10\angle 38.87° = 94.2\angle{-9.31°}^{\text{V}}$

c) V.R.

At no load, the rated voltage is the same as the secondary-side induced voltage, 120^{V}, and so the voltage regulation, that is, the change in the load voltage from no load to full load, is

$$VR = \frac{120 - 94.2}{94.2} = 0.2739 = 27.39\%.$$

d) Load power.

On the load side, apparent power can be calculated as voltage times current, $S = 94.2 \times 9.42 = 887.36^{\text{VA}}$, and real power, using the phase angle, is

$$P = S\cos(\theta) = 887.36 \times 0.8 = 709.89^{\text{W}}.$$

e) Efficiency.

Using provided formulas for copper and iron losses,

$$P_{cu} = I_2'^2(R_1 + R'_2) = 4.77^2(3) = 68.26^{\text{W}}$$

(remember that R refers only to the real part of the impedance, and neglects reactance), and

$$P_{iron} = \frac{V_1^2}{R_0} = \frac{240^2}{500} = 115.2^{\text{W}}.$$

Then efficiency is given by the total output power over the input power, as

$$\eta = \frac{709.89}{893.35} = 0.7946 = 79.46\%.$$